Supply Chain Analysis

1.Executive Summary

This diagnostic report presents a comprehensive analysis of the Supply Chain Analysis, covering product performance, logistics efficiency, defect management, and predictive risk metrics.

The dashboard spans three analytical pages — Products & Sales Relationship, Logistics Performance, and Prediction & Risk Analysis.

Collectively provides visibility into the end-to-end supply chain across three product categories: Skincare, Haircare, and Cosmetics.

The business operates across five Indian cities (Bangalore, Chennai, Delhi, Kolkata, Mumbai) and relies on five suppliers, three transport carriers, and three shipping routes. Key metrics as of the latest reporting period are summarized below:

16 Days Average Lead Time	**46K** Total Products Sold	**₹ 578K** Total Revenue	**₹ 47.27** Average Manufacturing Cost	**₹ 5.55** Average Shipping Cost	**2.28** Average Defect Rate

At a high level, the supply chain is generating solid revenue but facing challenges around lead time consistency, defect rates at specific suppliers, and shipping cost optimization. Predictive modelling further suggests significant revenue uplift potential if defect rates and lead times are systematically reduced.

2.Products & Sales Relationship

Figure 1: Products & Sales Relationship Dashboard

2.1 Products Sold by Category

Skincare leads the portfolio with 20.7K units sold, followed by Haircare at 13.6k units and Cosmetics at 11.8K units. The category dominance of Skincare in both volume and revenue positions it as the core revenue driver and demands priority in inventory and supply chain planning.

2.2 Revenue by Product Type

Area	Key Insight
Skincare	₹ 242K — highest revenue contributor (41.8% of total)
Haircare	₹ 174K — second largest category (30.1%)
Cosmetics	₹ 162K — lowest revenue but meaningful share (28%)
Total Revenue	₹ 578K across all categories

While Skincare dominates by volume and revenue, Cosmetics shows relatively strong revenue per unit compared to its volume share, indicating higher per-unit pricing or margin potential.

2.3 Product Profitability (Waterfall Chart)

The profitability waterfall chart reveals cumulative profit builds primarily through Skincare (₹ 239K increase) and Haircare (₹ 173K increase), with Cosmetics adding ₹ 160K to reach the total profit of ₹ 572K. The incremental nature of each category's contribution highlights the layered profitability structure.

2.4 Total Product Availability

Inventory levels show Skincare holding 2K units, Haircare 1K, and Cosmetics 1K in stock. This proportionate availability aligns with the sales volume pattern, though Skincare's higher sales velocity warrants continuous monitoring to prevent stockouts during peak demand periods.

2.5 Products Sold vs Revenue Scatter Plot

The scatter plot of Products Sold vs Revenue across SKUs reveals a wide distribution with no strong linear clustering at the higher volume end (800–1000 units). This suggests that a few high-volume SKUs are not necessarily the highest-revenue earners — pointing to a mix of premium and volume-driven SKUs that warrant distinct pricing and promotion strategies.

3.Logistics Performance

Figure 2: Logistics Performance Dashboard

3.1 Lead Time Analysis

Lead time performance varies significantly across suppliers. Supplier 1 records the highest total lead time of 504 days, followed by Supplier 2 at 453 days, while Supplier 3 shows the lowest at 215 days. This 134% variance between best and worst performers presents a meaningful opportunity for supplier negotiation and performance improvement.

Here Sum of Lead Times=Lead Times

Total Lead Times =Shipping Time + Manufacturing Lead Time

3.2 Routes & Average Lead Analysis

Route and carrier combination analysis identifies Route B with Carrier A as the most expensive pairing at 20 days, followed by Route B with Carrier B 16 days. On the other end, Route A with Carrier A is the most less lead time 10 days — a nearly 5x time difference compared to the average lead time of route.

Area	Key Insight
Route B + Carrier A	20 days
Route B + Carrier B	16 days
Route A + Carrier A	10 days
Less Lead Time	Route A + Carrier A

3.3 Shipping Cost by Carrier

Carrier B consistently commands the highest shipping costs across all product categories — 73 for Cosmetics, 106 for Haircare, and 58 for Skincare — making it the most expensive carrier overall. Carrier A, in contrast, shows the most competitive pricing, particularly for Cosmetics (29) and should be prioritized for cost-reduction strategies.

3.4 Defect Rate Analysis

Supplier-level defect rates reveal critical quality control gaps. Supplier 2 records the highest total defect rate of 51.98, largely driven by Haircare defects (24.53). Supplier 1 follows at 48.70, with a major contribution from Skincare (25.07). Supplier 3 demonstrates the best quality performance with a total of 36.99.

Area	Key Insight
Supplier 2	51.98 total defect rate — Highest Risk
Supplier 1	48.70 total defect rate — High Risk
Supplier 5	47.98 — Moderate Risk
Supplier 4	42.07 — Moderate Risk
Supplier 3	36.99 — Best Performer

3.5 Routes & Cost Analysis by Location

City-level cost analysis shows Bangalore as the most expensive hub across all three routes, with Route A costing ₹ 3.1K at that location. Chennai benefits from the most cost-efficient routing, making it the preferred gateway for cost-conscious distribution. Delhi shows significant cost variation across routes, suggesting room for route optimization.

3.6 Transportation Mode Distribution

Road transport dominates at 47.79 units, closely followed by Sea (50.18) and Air (51.58), with Rail at 47.93. The near-even distribution across modes reflects a multi-modal strategy, though the premium cost of Air transport should be evaluated against its lead time benefits for high-priority SKUs.

4. Prediction & Risk Analysis

Figure 3: Prediction & Risk Analysis Dashboard

4.1 Revenue Prediction (Linear Regression)

The Revenue Prediction scatter plot using Linear Regression on Price shows a relatively stable trend line around ₹ 5K–6K revenue across the price range of 0–100. The wide data scatter above the trendline indicates that factors beyond price alone — such as product type, marketing spend, and seasonal demand — are significant revenue drivers.

4.2 Defect Rate Prediction by Manufacturing Cost

The Defect Rate Prediction model reveals that defect rates are broadly distributed between 0 and 100 across manufacturing cost ranges of ₹ 0–5. Interestingly, there is no strong negative correlation between higher manufacturing cost and lower defects, suggesting that investment in quality controls and process improvement is more impactful than simply increasing production spend.

4.3 What-If: Revenue Uplift from Improvements

The dashboard's interactive What-If analysis calculates two powerful scenarios based on a 10% improvement:

- Defect Rate Reduction of 10%: Projected Revenue Uplift of ₹ 26.31K
- Lead Time Reduction of 10%: Projected Revenue Uplift of ₹ 197.31K

The lead time improvement scenario delivers nearly 7.5x more revenue benefit than defect reduction, underscoring that faster delivery is the single most commercially valuable operational improvement the business can make.

4.4 Supplier Revenue Forecasting

The What-If forecasting model presents three revenue scenarios per supplier:

Area	Key Insight
Supplier 1	Base: ₹ 158K | Defects Improve: ₹ 163K | Lead Time Improve: ₹ 204K
Supplier 2	Base: ₹ 125K | Defects Improve: ₹ 131K | Lead Time Improve: ₹ 172K
Supplier 5	Base: ₹ 110K | Defects Improve: ₹ 116K | Lead Time Improve: ₹ 150K
Supplier 3	Base: ₹ 98K | Defects Improve: ₹ 103K | Lead Time Improve: ₹ 137K
Supplier 4	Base: ₹ 86K | Defects Improve: ₹ 91K | Lead Time Improve: ₹ 113K

Supplier 1 holds the highest revenue potential across all scenarios, while Supplier 4 presents the lowest absolute upside — though even Supplier 4 could add ₹ 27K in revenue from lead time improvement alone.

4.5 Cost Efficiency by Location

Mumbai emerges as the most expensive operational location at ₹ 14.62 per unit cost, significantly above the average. Chennai, at ₹ 9.58, offers the most cost-efficient operating environment. This geographic cost spread of 53% between highest and lowest locations should inform warehouse siting and distribution center investment decisions.

4.6 Inspection Result Summary

Quality inspection outcomes across product categories show mixed pass/fail rates:

- Skincare: 13 Pass, 11 Fail, 16 Pending — highest pending backlog
- Haircare: 13 Pass, 6 Fail, 15 Pending
- Cosmetics: 10 Pass, 6 Fail, 10 Pending

The large number of pending inspections across all categories is a concern, as unresolved inspection status creates uncertainty in inventory availability and may mask actual defect rates.

4.7 Risk Distribution — Defect Rate × Lead Time

The risk scatter plot maps SKUs by defect rate (y-axis) against lead time (x-axis), revealing several high-risk outliers with both high defect rates (18–20%) and extended lead times (2.4–2.6 weeks). These represent the highest supply chain exposure and should be prioritized for immediate corrective action.

5. Strategic Recommendations

5.1 Immediate Actions (0–3 Months)

- **Address Supplier 2 and Supplier 1 quality issues — launch formal Supplier Quality Improvement Programs (SQIPs) with defined KPI targets**
  - Target: Reduce combined defect rate by at least 15% within 90 days
  - Focus on Haircare (Supplier 2: 24.53 defects) and Skincare (Supplier 1: 25.07 defects)
- **Resolve pending inspection backlog — prioritize Skincare's 16 pending items to unlock inventory clarity**
- **Restrict Route A + Carrier B usage to time-critical shipments only — shift routine volumes to Route C + Carrier A to reduce shipping cost**

5.2 Short-Term Actions (3–6 Months)

- **Invest in lead time reduction at Supplier 1 (504 units) — renegotiate SLAs or explore dual-sourcing**
  - A 10% lead time reduction alone can unlock ₹ 197K additional revenue
- **Shift more volume to Carrier A — its consistent cost-efficiency across all categories makes it the preferred logistics partner**
- **Expand Skincare inventory buffer given its high sales velocity — stock-out risk is material given current 2K availability**
- **Evaluate Mumbai operations costs — at ₹ 14.62/unit, explore whether consolidation or route restructuring can bring parity with Chennai (₹ 9.58/unit)**

5.3 Long-Term Strategy (6–12 Months)

- **Build a defect prediction model per SKU using manufacturing cost + supplier data to proactively flag quality risks before goods enter distribution**
- **Develop a supplier scorecard integrating lead time, defect rate, and cost efficiency — use it for quarterly supplier reviews and contract renegotiations**
- **Consider consolidating airfreight volumes with Rail transport for non-urgent Skincare shipments given comparable cost profiles and lower environmental impact**
- **Invest in regional distribution hubs in Chennai and Bangalore — lower operating costs and favorable route economics make them strategically attractive**

6. Conclusion

The Supply Chain Analysis dashboard provides a robust, multi-dimensional view of an operationally complex supply chain. The business is generating healthy revenue of ₹ 578K across three categories and five cities, but the data reveals clear opportunities to improve profitability through targeted operational improvements.

The single most impactful lever available is lead time reduction. At current volumes, a 10% improvement in lead times translates to ₹ 197K in additional revenue — nearly eight times the return from an equivalent defect rate improvement. This signals that customers and revenue are more sensitive to delivery speed than product quality issues, though both must be managed.

Supplier 1 and Supplier 2 require immediate quality management attention, while Route A with Carrier B should be deprioritized in favor of cost-efficient alternatives. Mumbai's elevated logistics costs merit a strategic review, and the large pending inspection backlog must be cleared to ensure supply chain transparency.

With focused execution on these recommendations, the business is well-positioned to significantly enhance supply chain efficiency, reduce operational costs, and drive meaningful revenue growth over the next 12 months.